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Quantity and Quality of Stem Wood Concerning Biofuel Production from Young *Paulownia* Species Planted in Tochigi, Japan

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Abstract

Paulownia forestry in Japan has traditionally focused on producing high-value solid wood for traditional household goods. In the present study, to explore the potential of a new *Paulownia* plantation forestry model aimed at meeting Japan's increasing demand for biofuels, the volume and weight of stems and the fuel characteristics of wood and bark were investigated for 15 four-year-old *Paulownia* trees planted in Tochigi, Japan. Fuel characteristics were also examined for nine common Japanese species as a reference. For *Paulownia* species, the bark had higher extractives, ash, and Klason lignin, lower contents of holocellulose and α-cellulose, and higher gross calorific value than wood. Potassium was the predominant inorganic component in both wood and bark in *Paulownia* species. Although the gross calorific value per unit volume of *Paulownia* species was the lowest among the tested nine Japanese species, the estimated annual gross calorific value per unit area was almost the same between *Paulownia* species and *Cryptomeria japonica* (the most common plantation softwood species in Japan) when the whole stem was used for biofuel production. Thus, *Paulownia* species might be used for biofuel production under short rotation cycles, even in lower bulk density of wood. On the other hand, wood from higher stem positions exhibited lower moisture content in *Paulownia* species. Therefore, drying costs might be reduced by using the lower stem portions for solid wood production, improving the overall economic efficiency of biofuel production.

Keywords Stem volume · Fuel characteristics · Bulk density · Chemical component · Calorific value · Element analysis

Introduction

Reducing greenhouse gas emissions from the atmosphere has become increasingly important in mitigating global warming. In Japan, fossil fuels, such as oil, coal, and natural gas, account for 83% of the country's total primary energy consumption and are the main contributors to greenhouse gas emissions [1]. To address this issue, the Feed-In Tariffs (FIT) system was introduced in 2012 to promote using renewable energy sources [1].

Among various renewable energy sources, woody biomass is regarded as an attractive biofuel option for power generation [2]. It includes logs, residual waste from the sawing process used in solid wood production, and other wood-based materials. In addition to being used as a biofuel, woody biomass also serves as a raw material for pulp, paper, and plywood production [3]. Consequently, when the demand for woody biomass increases in the pulp, paper, and plywood industries, the availability of biomass for biofuel production tends to decline, because biofuel prices are not consistently competitive. A stable fuel supply is a key challenge for promoting woody biomass power generation.

To meet the increasing demand for biofuels and reduce competition between energy and material uses of wood, plantations of fast-growing tree species have been established [3, 4]. These plantations typically consist of monocultures of a few selected species, such as *Acacia* spp., *Eucalyptus* spp., *Populus* spp., and *Salix* spp [2–4]. Such systems produce large amounts of biomass within a relatively short period. To optimize the use of woody biomass

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from these plantations for biofuel purposes, it is important to understand key wood properties, such as moisture content, calorific value, and ash content [3, 5]. For instance, high ash content can lead to fouling and slagging in biomass combustors and should be avoided [3, 6]. In contrast, a high calorific value enhances combustion efficiency [4]. These fuel characteristics are closely related to the chemical composition of the wood [2]. Consequently, the fuel characteristics as well as the wood quantity have been studied in fast-growing tree species e.g., [2–4].

The genus *Paulownia* comprises nine species [7] and is native to Northeast Asian regions, including Japan, China, Korea, and Taiwan [8, 9]. Several *Paulownia* species exhibit rapid growth, even in relatively cold climates [8]. Therefore, they have been widely planted as fast-growing tree species across Japan, other parts of Asia, Europe, the USA, and beyond [9, 10]. The wood of *Paulownia* species is generally light and soft [8]. Due to these wood characteristics, *Paulownia tomentosa* (Thunb.) Steud. has traditionally been planted in Japan to produce chest drawers, Japanese harps (*koto*), and various household goods [8]. Thus, *Paulownia* forestry site has been established in Japan [8]. In fact, the Forest Agency, Ministry of Agriculture, Forestry, and Fishery, Japan, classifies *Paulownia* as a forest byproduct rather than a timber-producing forestry species [11]. However, lifestyle changes in Japan have led to a decline in the use of traditional wooden products of *Paulownia tomentosa*, resulting in the reduction of traditional *Paulownia* wood production sites throughout Japan [10].

Since 2020, the *Paulownia* species has been planted as a fast-growing tree species on abandoned farmland in Mashiko and Motegi, Tochigi Prefecture, Japan, because the *Paulownia* species can be planted on farmland with appropriate management practices, such as fertilization, under the Agricultural Land Act in Japan. As of April 2025, approximately 12,000 trees have been planted in abandoned farmlands (in total, about 15 hectares). Using the fast-growing characteristics of the *Paulownia* species, the plantation is intended to produce wood resources but not produce the expensive traditional solid wood. One of the new wood products from the *Paulownia* plantation is considered as fuel wood, because the large quantity and production sustainability are required for fuel wood, due to the expansion of renewable energy use. This concept, thus, can be referred to as 'new *Paulownia* plantation forestry'. However, since these plantations are still at an early growth stage, information on tree growth performance and the biofuel characteristics of wood and bark in plantation-grown *Paulownia* has been still limited. Therefore, the accumulation of such scientific data is necessary to promote their establishment and sustainable management.

In the present study, the volume and weight of stems and fuel-related properties, including moisture content, bulk density, chemical components, gross calorific value, and element composition, were investigated in 15 four-year-old *Paulownia* trees planted in Tochigi, Japan. Based on these results, the potential of *Paulownia* trees grown on abandoned farmland as a woody biomass resource for biofuel production was evaluated.

Materials and Methods

Materials

The plantation (0.89 ha) of *Paulownia* species was established in Mashiko, Tochigi Prefecture, Japan (N36º29'01'', E140º08'09'') in July 2020, using seedlings with an initial spacing of 3 by 3 m (Fig. 1). Following planting, annual treatments including fertilization, weed control, disease and insects control, and pruning were conducted. Although no thinning was performed, severely bent or decayed trees were cut at the base, resulting in approximately 5% of the total planted trees being removed from after establishing the plantation to March 2024. Of the 692 remaining trees in the plantation, stem diameter at 1.3 m above the ground and tree height were measured in 288 trees at that time (Fig. 1). Then, 15 trees with stem diameters close to the mean and without any visible defects (e.g., decay) were selected for sampling. The mean ± standard deviation of stem diameter and tree height in the selected trees were 15.6 ± 1.2 cm and 9.7 ± 0.8 m, respectively. Disks (3 cm in thickness) were collected at 0.3 m and 2 m intervals from 1.3 to 9.3 m along the stem after the trees were harvested. One-meter-long logs were obtained from 0.3 to 1.3 m above the ground, and two-meter-long logs were taken sequentially upward from 1.3 m toward the tree top. In addition, top portions of the stems were collected. The green weight of logs and remaining stems was measured using an electronic balance (HCSA01, HELIX, China).

To compare biofuel characteristics, nine additional Japanese tree species were analyzed as reference: *Cryptomeria japonica* (L.f.) D.Don, *Chamaecyparis obtusa* (Siebold et Zucc.) Endl., *Larix kaempferi* (Lamb.) Carrière, *Pinus densiflora* Siebold et Zucc., *Quercus serrata* Murray, *Cerasus jamasakura* (Siebold ex Koidz.) H.Ohba, *Castanea crenata* Siebold et Zucc., *Magnolia obovata* Thunb., and *Paulownia tomentosa*. Except for *Paulownia tomentosa*, 3-cm-thick sample disks were collected at 1.3 m above the ground from trees growing in a University Forest, School of Agriculture, Utsunomiya University (N36º46'25'', E139º49'16''). For *Paulownia tomentosa*, a 10-cm-thick disk was obtained at 0.4 m above the ground from a tree planted in a private forest in Kanayama, Fukushima, Japan (N37º21'62'', E139º26'84') [12].

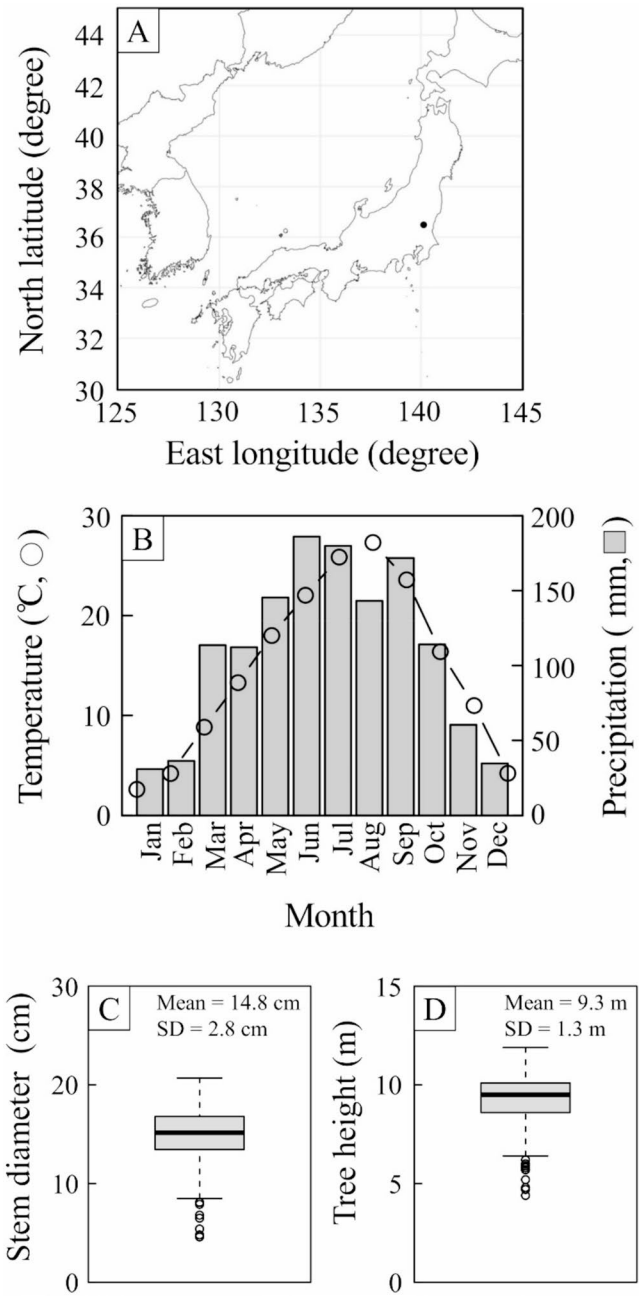


Fig. 1 Map (**A**), climatic conditions (**B**) of a *Paulownia* plantation, and growth characteristics (stem diameter at 1.3 m above the ground [**C**] and tree height [**D**]) of 288 sample trees. Note: SD, standard deviation. Bars in B indicate the precipitation. Solid line with circles in B represents the temperature

Stem Analysis

The transverse surfaces of the collected *Paulownia* disks were sanded. Then, the digital images (resolution = 600 dpi) of the transverse section were obtained using a flatbed scanner (CanoScan LiDE 400, Canon, Tokyo, Japan). Annual ring width was measured in four radial directions on each disk using ImageJ software (version 1.54 g) [13]. The mean value of the four measurements was calculated and used as the representative annual ring width for each sampling height in each tree.

Moisture Content and Bulk Density

Fan-shaped specimens (with a center angle of 30° and without defects such as knots and decay) were prepared from disks collected at various heights. These specimens were further sectioned into small samples in each annual ring. The weight and volume of each small specimen were measured in the green conditions. Green volume was determined using the water displacement method. After volume measurement, the specimens were oven-dried at 105 °C until a constant weight was reached. Moisture content (MC) and bulk density (BD) were calculated using the following equations:

$$\mathrm{MC}(\%) = (Wg - Wo)/Wg \times 100$$

$$\mathrm{BD}\ (\mathrm{kg\ m^{-3}}) = Wo\ /\ Vg$$

where *Wg* is the green weight, *Wo* is the oven-dried weight, and *Vg* is the green volume.

For comparison, the bulk density of other Japanese species was determined using small wood blocks (10 by 10 by 10 cm) collected from the disks. As these blocks had been previously air-dried, they were dipped into tap water for one week. Bulk density was calculated using the same method as for specimens of *Paulownia* sp.

Amounts of Wood Chemical Components

The following chemical components were determined for wood (xylem) and bark of *Paulownia* species using the ordinal method [14]: hot-water extracts, 1% NaOH extracts, ethanol-toluene extracts, ash, Klason lignin, holocellulose, and α-cellulose. Wood and bark powders were prepared using several types of mills and subsequently sieved to obtain particles within the 42 to 82 mesh range for chemical component analysis. To compare the biofuel characteristics, the ash content of the nine Japanese species was also determined using the same method as that used for the *Paulownia* species.

Calorific Value

The gross calorific value of wood and bark was determined for *Paulownia* species and nine Japanese species. Oven-dried samples (0.2 g for wood samples and 0.4 to 0.6 g for bark samples) were used for measurement with a bomb calorimeter (C200, IKA, Staufen, Germany).

Element Analysis for Wood and Bark

Wood specimens (15 [R] by 15 [T] by 3 to 15 [L] mm) and bark specimens (15 by 15 mm in surface section) were prepared from 15 *Paulownia* trees. Prior to analysis, all samples were dried at 105 °C in a laboratory oven until reaching constant weight. Element analysis was performed using an X-ray fluorescence analyzer (ZSX PrimusII, Rigaku, Tokyo, Japan). The mass percentages of the following elements were determined: Na, Mg, Al, P, S, Cl, K, Ca, Cr, Fe, Cu, Zn, As, Cd, Hg, and Pb.

Statistical Analysis

All statistical analyses were conducted using R software (version 4.4.2) [15]. To develop a taper equation, half of the butt-end diameter was considered as a log radius. Following the methodology proposed by Salas-Eljatib [16], a taper equation was developed using a sigmoidal model fitted to stem diameter data at multiple height positions in coniferous and broad-leaved tree species [16]. In the present study, a logistic function, one form of the sigmoidal model, was applied, with height position above the ground as the explanatory valuable and log diameter as the response variable. Stem volume was estimated as the volume of revolution, calculated by integrating the taper equation from the base (butt) to the top end of each log.

To assess longitudinal variations in moisture content and bulk density, exponential regression models were fitted to the measured data. These models were used to estimate moisture content and bulk density at heights of 1.3, 3.3, 5.3, 7.3, and 9.0 m above the ground. Using the estimated bulk density and wood volume, oven-dry weight was calculated at 2-m intervals from 0.3 m (base) to 9.7 m (top, corresponding to the mean tree height). The gross calorific value per unit volume was determined by multiplying bulk density by the gross calorific value of wood.

A *t*-test was used to evaluate differences in the amounts of chemical components and calorific value between wood and bark. In addition, Pearson's correlation coefficients were calculated to assess the relationships between the amounts of wood chemical components and calorific value.

Results

The results of the stem analysis and the obtained taper equation are presented in Figs. 2 and 3. Figure 4 shows the longitudinal variations of moisture content and bulk density along with their regression curves. A continuous pith cavity was observed from the base to the top of the stem (Fig. 2). In addition, the tree height reached 3.3, 7.3, and 9.7 m (top) above the ground at the age of one, two, and three years, respectively (Fig. 2).

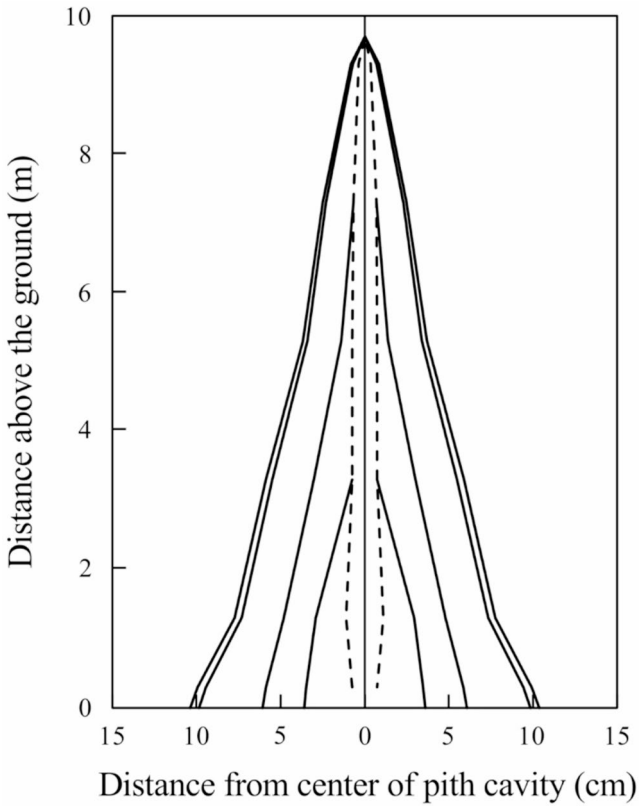


Fig. 2 Results of stem analysis. Note: Each solid line represents the mean value of 15 trees at each age. Dotted lines indicate the pith cavity

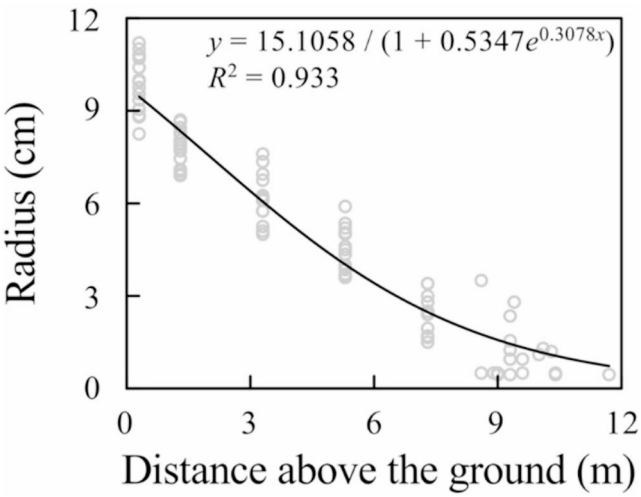


Fig. 3 Taper equations of 4-year-old *Paulownia* species. Note: R^2, Coefficient of determination. Circles in this figure indicate the obtained data from 15 trees. The regression curve is based on a logistic function

Moisture content, bulk density, and stem volume estimated using the regression formulas in Figs. 3 and 4 are summarized in Table 1. The mean values of the estimated moisture content and bulk density were 58.2% and 255 kg m^{-3}, respectively. The stem volume from 0.3 to 2.3 m

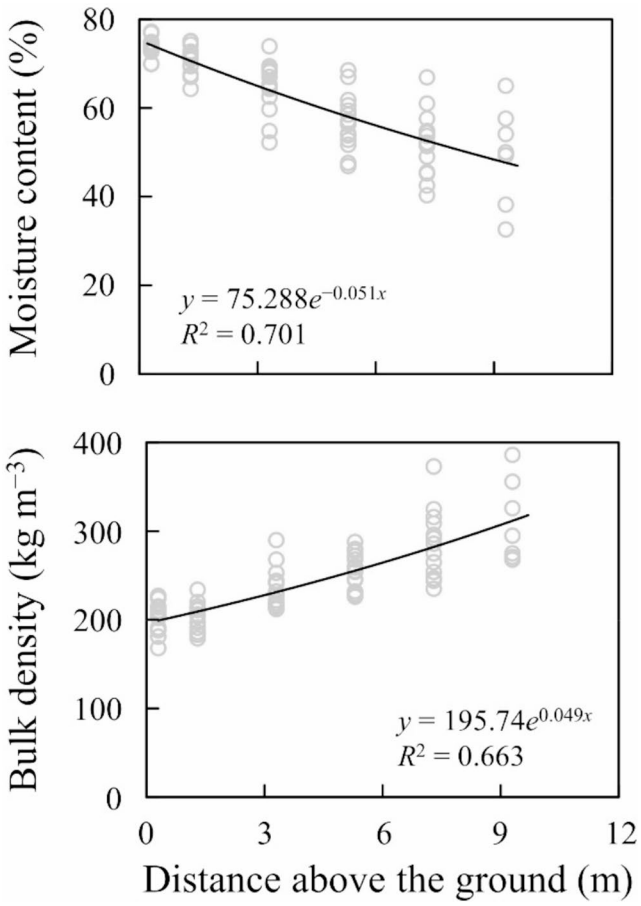


Fig. 4 Longitudinal variations of moisture content and bulk density in *Paulownia* species trees. Note: R^2, Coefficient of determination. Circles in the figures indicate the obtained data from 15 trees. The regression curves are based on an exponential function

Table 1 Estimated volume and oven-dry weight in each height position

Distance above the ground (m)	MC (%)	BD (kg m^{-3})	WV (m^3)	ODW (kg)
0.3–2.3	70.5	209	0.0439	9.2
2.3–4.3	63.6	230	0.0234	5.4
4.3–6.3	57.5	254	0.0104	2.6
6.3–8.3	51.9	280	0.0040	1.1
8.3–9.7	47.6	304	0.0011	0.3
Mean	58.2	255	–	–
Total	–	–	0.0828	18.6

MC moisture content, *BD* bulk density. *WV* wood volume, *ODW* oven-dry weight. Moisture content and bulk density were estimated by the regression formula in Fig. 4 at the height positions of 1.3, 3.3, 5.3, 7,3, and 9.0 m above the ground. Wood volume was calculated using the regression formula in Fig. 3. Oven-dry weight was estimated by the estimated bulk density at each height position, multiplying the estimated wood volume

above the ground was 0.0439 m^3. The estimated oven-dry weights were 9.2 kg, 5.4 kg, and 2.6 kg for logs at height intervals of 0.3–2.3 m, 2.3–4.3 m, and 4.3–6.3 m, respectively. The total stem volume and estimated oven-dry weight from the base to the tree top were 0.0828 m^3 and 18.6 kg, respectively.

Amounts of chemical components and gross calorific values in wood and bark are listed in Table 2. Mean values of chemical components in wood were 12.8% in hot-water extracts, 24.1% in 1% NaOH extracts, 6.0% in organic solvent extracts, 1.7% in ash, 20.5% in Klason lignin, 76.1% in holocellulose, and 40.5% in α-cellulose. The corresponding values for bark were 16.8%, 41.5%, 7.7%, 4.5%, 41.5%, 57.8% and 29.2%, respectively. The gross calorific values were 18.9 MJ kg^{-1} for wood and 20.5 MJ kg^{-1} for bark. Significant differences were found between wood and bark in the amount of chemical components and gross calorific value.

The results of the elemental analysis are shown in Table 3. Except for carbon (C) and oxygen (O), potassium (K) was the most abundant element, with a concentration of 0.4 mass% in wood and 1.5 mass% in bark.

Table 4 presents the correlation coefficients between chemical components and gross calorific value. No significant correlations were found in wood. However, significant correlations were observed in bark between 1% NaOH extracts, ash, Klason lignin or α-cellulose and calorific value.

Tables 5 and 6 shows the comparison of gross calorific value and ash content in wood and bark of *Paulownia* species with nine Japanese species. The variation in gross calorific values per gram of oven-dried wood and bark among the nine species was relatively small, ranging from 18.5 MJ kg^{-1} (*Castanea crenata*) to 20.2 MJ kg^{-1} (*Cryptomeria japonica* and *Pinus densiflora*) for wood, and from 19.7 MJ kg^{-1} (*M. obovata*) to 21.9 MJ kg^{-1} (*Quercus serrata*) for bark (Table 5). Compared to other Japanese species, the gross calorific values in *Paulownia* species showed lower values in wood (18.9 MJ kg^{-1}) and moderate values in bark (20.5 MJ kg^{-1}) (Table 5). In contrast, the gross calorific value per unit volume of *Paulownia* species (4.0 GJ m^{-3}) was the lowest among the species, corresponding to about 40% of the highest value (11.1 GJ m^{-3}) recorded in *Quercus serrata*, and about 70% of the value (6.1 GJ m^{-3}) in *Cryptomeria japonica*, which is the major plantation tree species in Japan (Table 5). Ash content in the wood of *Paulownia* species (1.7%) was higher than that of the nine Japanese species (Table 6). Bark also showed relatively high ash content, although similar levels were observed in other hardwoods, ranging from 2.9% (*Paulownia tomentosa*) to 4.5% (*Magnolia obovata*) (Table 6).

Discussion

Wood Volume

Kumakura [8] examined the growth characteristics of *Paulownia tomentosa* trees planted at a historically renowned traditional *Paulownia* forestry site in Japan to develop an

Table 2 Chemical components and gross calorific value of wood and bark

Property	Wood				Bark				*t*-value	*p*-value
	Mean	SD	Min	Max	Mean	SD	Min	Max		
HWE (%)	12.8	2.4	7.5	16.7	16.8	2.4	11.4	23.7	−7.176	< 0.001
AE (%)	24.1	2.2	19.5	29.6	41.5	2.4	33.3	49.6	−29.172	< 0.001
OSE (%)	6.0	3.4	1.5	13.2	7.7	1.6	3.7	12.3	−2.732	0.007
Ash (%)	1.7	0.2	1.1	2.1	4.5	0.6	3.3	20.7	−28.213	< 0.001
KL (%)	20.5	1.5	15.7	27.3	41.5	2.8	34.9	48.1	−31.607	< 0.001
HC (%)	76.1	1.9	68.9	79.4	57.8	2.2	52.4	64.4	34.300	< 0.001
AC (%)	40.5	1.6	34.8	43.2	29.2	1.7	24.0	38.3	22.700	< 0.001
GCV (MJ kg^{-1})	18.9	0.1	18.3	19.3	20.5	0.4	19.7	21.5	−22.896	< 0.001

Number of samples = 15 individual trees. *SD* standard deviation, *Min* minimum, *Max* maximum, *HWE* hot-water extracts, *AE* 1% NaOH extracts, *OSE* organic solvent (ethanol-toluene) extracts, *KL* Klason lignin, *HC* holocellulose, *AC* α-cellulose, *GCV* gross calorific value. The *t*- and *p*-values were obtained by *t*-test to detect the difference in each chemical component between wood and bark

Table 3 Results of X-ray fluorescence analysis in wood and bark of *Paulownia* species

Element	Mass (%)			
	Wood		Bark	
	Mean	SD	Mean	SD
C	52.5	0.6	54.0	0.9
O	46.9	0.5	43.2	1.0
Na	<0.1	–	ND	–
Mg	<0.1	<0.1	0.2	<0.1
Al	<0.1	<0.1	<0.1	<0.1
Si	<0.1	<0.1	<0.1	<0.1
P	<0.1	<0.1	0.2	<0.1
S	<0.1	<0.1	0.1	<0.1
Cl	<0.1	<0.1	<0.1	<0.1
K	0.4	0.1	1.5	0.7
Ca	<0.1	<0.1	0.8	0.2
Fe	<0.1	<0.1	<0.1	<0.1
Cu	<0.1	<0.1	<0.1	<0.1
Zn	<0.1	<0.1	<0.1	<0.1
Rb	ND	–	<0.1	–

Number of samples = 15 individual trees, *SD* standard deviation, *ND* not detected

Table 4 Correlation coefficients between wood chemical components and calorific value for wood and bark

Property	Wood		Bark	
	r	*p*-value	*r*	*p*-value
Hot-water extracts	−0.160	0.568	−0.184	0.511
1% NaOH extracts	−0.283	0.307	−0.543	0.037
Ethanol-toluene extracts	0.096	0.734	0.300	0.278
Ash	0.410	0.129	−0.696	0.004
Klason lignin	−0.225	0.421	0.628	0.012
Holocellulose	0.426	0.114	0.309	0.263
α-cellulose	0.020	0.944	0.565	0.028

Number of samples = 15 individual trees

empirical yield table. According to this yield table, under optimal site conditions and an initial spacing of 3.65 by 3.65 m, the stem diameter at 1.2 m above the ground and the wood volume per tree were 13.6 cm and 0.025 m^3, respectively, at 4 years old. In the present study, the 4-year-old *Paulownia* species showed a stem diameter of 15.6 cm at 1.3 m above the ground and a wood volume of 0.083 m^3 (Table 1). This wood volume was approximately three times greater than that reported by Kumakura [8] despite the narrower initial spacing of 3 × 3 m used in the present study. These findings suggest that the *Paulownia* species examined in the present study exhibited superior radial and height growth compared to the traditionally planted *Paulownia tomentosa* in Japan.

Moisture Content of Wood

Moisture content is a key quality indicator of biofuel because it is closely related to the calorific values and combustion performance [3, 5]. According to the grading of wood chips defined by the Japan Woody Bioenergy Association [17], wood chips with more than 55% moisture content are unsuitable as fuel. In the present study, the estimated moisture content was either above or close to 55% at all height positions (Table 1; Fig. 4). Therefore, to produce high-quality biofuels from the wood of *Paulownia* species, a drying process must be incorporated during the production stage, either in log or chip form. In addition, moisture content decreased with increasing height along the stem (Fig. 4), indicating that wood harvested from upper stem positions might have an advantage in reducing drying costs and enhancing fuel quality.

Amounts of Wood Chemical Components and Calorific Values

The chemical components of wood in *Paulownia tomentosa* have been reported by the Japan Wood Technological Association [18] as follows: organic-solvent (ethanol-benzene) extractives, 4%; Klason lignin, 22%; *α*-cellulose, 39%; and ash, 0.4%. In addition, Senelwa and Sims [4] reported that the organic-solvent (ethanol-benzene) extractive contents and ash of wood and bark in 3-year-old *Paulownia tomentosa* were 8.8% and 19.1%, and 1.2% and 5.1%, respectively. With few exceptions, the chemical components of the

Table 5 Comparison of gross calorific value in wood and bark of *Paulownia* species with the nine Japanese species

	Species	Wood			Bark			Bulk density (kg m^{-3})	Estimated gross calorific value per unit volume	
		Mean (MJ kg^{-1})	SD (MJ kg^{-1})	Ratio	Mean (MJ kg^{-1})	SD (MJ kg^{-1})	Ratio		(GJ m^{-3})	Ratio
Softwood	*Cryptomeria japonica*	20.2	0.2	1.07	20.1	0.1	0.98	302	6.1	1.53
	Chamaecyparis obtusa	19.4	0.2	1.03	20.6	0.4	1.00	358	6.9	1.73
	Larix kaempferi	19.1	0.3	1.01	21.6	0.0	1.05	473	9.0	2.25
	Pinus densiflora	20.2	0.1	1.07	20.5	0.1	1.00	525	10.6	2.65
Hardwood	*Quercus serrata*	19.0	0.0	1.01	21.9	0.2	1.07	584	11.1	2.78
	Cerasus jamasakura	18.9	0.1	1.00	19.9	0.1	0.97	520	9.8	2.45
	Castanea crenata	18.5	0.2	0.98	21.1	0.2	1.03	410	7.6	1.90
	Magnolia obovata	19.5	0.1	1.03	19.7	0.0	0.96	350	6.8	1.70
	Paulownia tomentosa	19.1	0.1	1.01	20.9	0.1	1.02	239	4.6	1.15
	Paulownia species	18.9	0.1	–	20.5	0.4	–	209	4.0	–

SD standard deviation. Data is the mean and standard deviation of 15 trees in the *Paulownia* species and three replicates from an individual tree per species for the nine Japanese species. The ratio was calculated by mean data in each species to the *Paulownia* species. The estimated gross calorific value per unit volume was calculated by the bulk density value multiplying the actual gross calorific value of wood

Table 6 Comparison of Ash content in wood and bark of *Paulownia* species with the nine Japanese species

	Species	Wood			Bark		
		Mean (%)	SD (%)	Ratio	Mean (%)	SD (%)	Ratio
Softwood	*Cryptomeria japonica*	0.4	<0.1	0.24	0.5	0.2	0.11
	Chamaecyparis obtusa	0.1	0.0	0.06	0.8	<0.1	0.18
	Larix kaempferi	0.1	0.0	0.06	0.8	<0.1	0.18
	Pinus densiflora	0.3	<0.1	0.18	2.8	0.1	0.62
Hardwood	*Quercus serrata*	0.3	<0.1	0.18	3.5	0.4	0.78
	Cerasus jamasakura	0.3	<0.1	0.18	3.1	0.4	0.69
	Castanea crenata	0.2	<0.1	0.12	4.2	0.3	0.93
	Magnolia obovata	0.3	<0.1	0.18	4.5	0.1	1.00
	Paulownia tomentosa	0.1	<0.1	0.06	2.9	0.1	0.64
	Paulownia species	1.7	0.2	–	4.5	0.6	–

SD standard deviation. Data is the mean and standard deviation of 15 trees in the *Paulownia* species and three replicates from an individual tree per species for the nine Japanese species. The ratio was calculated by mean data in each species to the *Paulownia* species

Paulownia wood used in the present study (Table 2) were within the range of those previously reported [4, 18].

Differences in chemical components and gross calorific values between wood and bark were examined in the *Paulownia* species used in the present study. Compared to wood, bark exhibited (1) higher contents of extractives, ash, and Klason lignin; (2) lower contents of holocellulose and α-cellulose; and (3) higher gross calorific value (Table 2). Previous research has indicated a positive relationship between carbon content and the gross calorific value of biomass [5, 19]. Because lignin has much higher carbon content compared to polysaccharides, such as cellulose and hemicellulose, resulting that lignin contributes significantly to the gross calorific value [5, 19]. Thus, an increase in total lignin content contributes to a high calorific value of biomass [5, 19]. In Table 4, a significant positive correlation was also found between Klason lignin and gross calorific value in bark ($r = 0.628$, $p < 0.012$), indicating that the lignin contents of the bark influenced the gross calorific values of *Paulownia* bark. However, Klason lignin in bark contained not only lignin but also extractives, such as phenolic compounds and suberin. Therefore, these phenolic compounds, suberin, and other chemical compounds also might contribute to the gross calorific value of bark.

Comparison of biofuel characteristics of Paulownia species with the nine Japanese species

In this study, the biofuel characteristics of *Paulownia* species were compared with those of nine major Japanese species to evaluate their potential as renewable fuel resources. Although the *Paulownia* wood has a lower calorific value per unit volume compared with the other species, its annual increment of wood volume is usually greater than those of these Japanese species. Thus, *Paulownia* species can be utilized for biofuel production if short rotation cycles are applied, despite their low calorific value per unit volume.

Table 7 Comparison of stem volume, oven-dry wright, and calorific value in the different utilization scenarios

Scenario	Final products		Stem volume		Oven-dry weight		Annual value	
	Height position	Products	Per tree (m^3)	Per ha (m^3)	Per tree (kg)	Per ha (t)[*1]	Increment ($t\ ha^{-1}\ yr^{-1}$)	Calorific value ($GJ\ ha^{-1}\ yr^{-1}$) [*2]
1	Whole	Biofuel	0.0828	87.35	18.6	19.6	4.9	93
2	0.2 to 2.0 m	Solid wood	0.0439	46.31	9.2	9.7	2.4	–
	2.0 m to tree top	Biofuel	0.0389	41.04	9.4	9.9	2.5	47
3	0.2 to 2.0 m	Solid wood	0.0439	46.31	9.2	9.7	2.4	–
	2.0 to 4.0 m	Solid wood	0.0234	24.69	5.4	5.7	1.4	–
	4.0 m to tree top	Biofuel	0.0155	16.35	4.0	4.2	1.1	20
48-year-old *Cryptomeria japonica* stem[*3]			0.375	858.00	121.1[*4]	277.1	5.0	101

[*1], Values were calculated by measured value multiplying 1055 trees ha^{-1} in stand density (initial planting density [1111 trees ha^{-1}, 3 × 3 m spacing] × the survival rate of 0.95). [*2], Values were calculated using mean calorific value of *Paulownia* species (18.9 MJ kg^{-1}) and *Cryptomeria japonica* (20.2 MJ kg^{-1}) listed in Table 5. [*3], Data was cited from Matsue [20] (initial planting density = 3000 trees ha^{-1}, standing tree density = 2288 trees ha^{-1}). [*4], Values were calculated using bulk density value of 323 kg m^{-3} [21] obtained the same study site with Matsue [20]. –, data is not shown because the final products are not biofuel

As shown in Table 6, the ash content in *Paulownia* wood and bark is relatively high compared with other species. According to the grading criteria of wood chips defined by the Japan Woody Bioenergy Association [17], wood chips are divided into four classes by ash content: class 1 (< 1.0%), class 2 (< 1.5%), class 3 (< 3.0%), and class 4 (< 5.0%). Based on these criteria, the *Paulownia* species in the present study were graded as class 3 for wood and class 4 for bark.

Potassium is well known to promote clinker formation during combustion because it forms low-melting-point compounds that fuse and accumulate within furnaces [3, 6]. The clinker reduces combustion efficiency and increases maintenance requirements [3, 6]. In *Paulownia* species used in this study, the predominant inorganic component was potassium in both wood and bark (Table 3). Therefore, when using *Paulownia* species as a biofuel, its high potassium content should be considered, and appropriate strategies might be necessary to minimize clinker generation.

Concluding Remarks

In the present study, wood volume and biofuel characteristics of four-year-old *Paulownia* species planted in Tochigi, Japan, were evaluated for potential use as biofuel. Based on the results, three scenarios for producing biofuel as well as solid wood from short-rotation *Paulownia* forestry in Japan were prepared. In each scenario, the final products of the wood from stems are as follows:

Scenario 1: Biofuel production from the whole stem.
Scenario 2: Solid wood production from the first 2-m-log (harvested from 2 to 4 m above the ground); the remaining portion used for biofuel.
Scenario 3: Solid wood production from the first and second 2-m-logs (harvested from 2 to 6 m above the ground); the remaining portion used for biofuel.

Table 7 represents the estimated wood volume, oven-dry weight, and gross calorific value of stem wood in each scenario. The estimated wood volume for biofuel production per hectare was 87.35 m^3 in Scenario 1, 41.04 m^3 for Scenario 2 (about half of Scenario 1), and 16.35 m^3 in Scenario 3 (about one-fifth of Scenario 1). The oven-dry weight per unit area for biofuel was 19.6, 9.9, and 4.2 t ha^{-1} for Scenarios 1, 2, and 3, respectively. The oven-dry weight and gross calorific value per unit area were estimated: 4.9 t ha^{-1} yr^{-1} and 93 GJ ha^{-1} yr^{-1} for Scenario 1, 2.5 t ha^{-1} yr^{-1} and 47 GJ ha^{-1} yr^{-1} for Scenario 2, and 1.1 t ha^{-1} yr^{-1} and 20 GJ ha^{-1} yr^{-1} for Scenario 3, respectively.

To evaluate these estimated values, the same factors were assessed for *Cryptomeria japonica*, the most widely planted species in Japan, using the standard harvesting age (approximately 50 years; 20). The bulk density value of *Cryptomeria japonica* was cited from a reference [21]. The wood volume per unit area of *Cryptomeria japonica* was estimated at 858.00 m^3 ha^{-1}, which is about nine times higher than that of the 4-year-old *Paulownia* species. The oven-dry weight per unit area showed a similar tendency: 237.8 t ha^{-1} in *Cryptomeria japonica*, approximately six times greater than that of *Paulownia* species. On the other hand, the oven-dry weight per unit area per year in *Paulownia* species (4.9 t ha^{-1} yr^{-1}) was almost the same as that in *Cryptomeria japonica* (5.0 t ha^{-1} yr^{-1}). A similar result was obtained in the annual gross caloric value per unit area. These results suggest that *Paulownia* species can achieve comparable annual energy yields to *Cryptomeria japonica*.

The moisture content of biofuel should be low. In the present study, wood from higher stem positions exhibited lower moisture content (Table 1; Fig. 4). Therefore, utilizing the lower portions of the stem for solid wood production, as in Scenarios 2 and 3, can reduce drying costs and enhance the economic efficiency of biofuel production. In addition, biofuel does not necessarily yield a higher

market value than solid wood. Thus, a combination of biofuel production and solid wood production can contribute to more sustainable biofuel production by improving overall profitability.

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Data Availability The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Competing Interests The authors declare no competing interests.

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